

Amirhossein Kazerouni

✉ amirhossein477@gmail.com | 🏠 amirhossein-kz.github.io/ | 📧 amirhossein-kz | 🌐 amirhossein477 | 🎓 Amirhossein Kazerouni

Education

University of Toronto (U of T)

Toronto, Canada

PhD in Computer Science – Advised by Prof. Babak Taati and Prof. Michael Brudno

Jan. 2024 - present

- **Courses:** Probabilistic Machine Learning, Generative AI for Images, Physics-Informed Neural Representations for Visual Computing, Physics-Based Animation, Mathematical Foundations of Algorithmic Fairness

Iran University of Science and Technology (IUST)

Tehran, Iran

B.S. in Electrical Engineering

Sep. 2017 - Feb. 2022

- **GPA:** 17.95/20 (3.85/4)
- **Thesis topic:** Design, Simulation, and Construction of an Autonomous Vehicle with Environment Perception, Planning, and Control Capabilities. (**Thesis Grade: 20/20**) — **Supervisor:** Dr. Saeed Shamaghdari, Associate Professor at IUST

Research Interests

- Scene Understanding and 3D Vision
- Scalable and Efficient Generative Modeling
- Image Restoration
- Diffusion Models
- Implicit Neural Representations
- Vision-Language Models

Research Experience

Research Intern, Samsung Toronto AI Center

Toronto, Canada

Supervisors: Dr. Alex Levinstein, Dr. Iqbal Mohamed, Prof. Konstantinos G. Derpanis

May 2025 – Present

- Developed Face2Scene, a two-stage diffusion-based scene restoration framework that extracts degradation cues from a restored-degraded face pair and restores the full image in a single step.
- Developed a deployment-friendly policy that prunes redundant visual tokens to save prefill compute and spend it on more self-consistency reasoning rollouts, consistently boosting VLM accuracy at matched inference cost.

Graduate Research Assistant, University of Toronto & Vector Institute

Toronto, Canada

Supervisors: Prof. Babak Taati and Prof. Michael Brudno

Jan. 2024 – Present

- Developing generative image and video models for 4D scene generation from video, with a focus on scene understanding, and task-agnostic representation learning.

Remote Research Assistant, RWTH Aachen University

Aachen, Germany

Supervisors: Prof. Dorit Merhof and Dr. Reza Azad

2022 – 2024

- Developed Transformer- and diffusion-based models for medical image segmentation, including multi-scale, self-supervised, and boundary-aware architectures for robust dense prediction.
- Developed prior-conditioned INRs that inject prior knowledge into sinusoidal activations for solving inverse problems.

Remote Machine Learning and Computer Vision Researcher, DGSculptor

Montreal, Canada

Supervisor: Dr. Alireza Morsali

2021 – 2023

- Co-developed an ensemble architecture for implicit neural representations to improve modeling flexibility and representation quality.

Computer Vision Researcher, IUST AI and Control Lab

Tehran, Iran

Supervisor: Prof. Saeed Shamaghdari

2019 - 2021

- Developed a fusion-based multi-stage approach that performs 3D shape analysis on vehicles to measure the amount of load protrusion.

Selected Publications

1. **Face2Scene: Using Facial Degradation as an Oracle for Diffusion-Based Scene Restoration**
Amirhossein Kazerouni, Maitreya Suin, Tristan Ty Aumentado-Armstrong, Sina Honari, Amanpreet Walia, Iqbal Mohamed, Konstantinos G. Derpanis, Babak Taati, Alex Levinstein
Published in **IEEE/CVF CVPR (2026)** (Samsung Research Toronto AI Center Internship) ([Paper](#), [Project Page](#))
2. **LIFT: Latent Implicit Functions for Task- and Data-Agnostic Encoding**
Amirhossein Kazerouni, Soroush Mehraban, Michael Brudno, Babak Taati
Published in **IEEE/CVF ICCV (2025)** ([Paper](#), [Project Page](#))
3. **SUM: Saliency Unification through Mamba for Visual Attention Modeling**
Alireza Hosseini*, **Amirhossein Kazerouni***, Saeed Akhavan, Michael Brudno, Babak Taati
Published in **IEEE/CVF WACV 2025** (Oral presentation) ([Paper](#), [GitHub](#))
4. **INCOCODE: Implicit Neural Conditioning with Prior Knowledge Embeddings**
Amirhossein Kazerouni, Reza Azad, Alireza Hosseini, Dorit Merhof, Ulas Bagci
Published in **IEEE/CVF WACV 2024** ([Paper](#), [GitHub](#), [Project Page](#))

Additional Publications

Conference / Workshop Papers

1. **PriLoRA: Prior-Conditioned Low-Rank Adapters for Medical Vision Models**
Amirhossein Kazerooni*, Sanaz Karimijafarbigloo*, Ali Aghayari*, Ali Ghasemzadeh Khorzoughi*, Reza Azad, Wojciech Samek, Dorit Merhof, Michael Brudno, Babak Taati
Accepted in **MICCAI 2026**
2. **Probabilistic Multi-Rater Segmentation via Style-Aware Boundary Conditioning**
Sanaz Karimijafarbigloo, **Amirhossein Kazerooni**, Alireza Kheirkhah, Sergey Protserov, Reza Azad, Michael Brudno, Babak Taati, Wojciech Samek, Dorit Merhof
Accepted in **MICCAI 2026**
3. **A Unified Theory of Sinusoidal Activation Families for Implicit Neural Representations**
Alireza Morsali, MohammadJavad Vaez, Mohammad Hossein Soltani, **Amirhossein Kazerooni**, Babak Taati, Morteza Mohammad-Noori
Published in **Transactions on Machine Learning Research (TMLR) (2026)** ([Paper](#), [GitHub](#), [Project Page](#))
4. **FuseNet: Self-Supervised Dual-Path Network for Medical Image Segmentation**
Amirhossein Kazerooni, Sanaz Karimijafarbigloo, Reza Azad, Yury Velichko, Ulas Bagci, Dorit Merhof
Published in **ISBI 2024** ([Paper](#), [GitHub](#))
5. **Beyond Self-Attention: Deformable Large Kernel Attention for Medical Image Segmentation**
Reza Azad, Leon Niggemeier, Michael Huttemann, **Amirhossein Kazerooni**, Ehsan Kh. Aghdam, Yury Velichko, Ulas Bagci, Dorit Merhof
Published in **IEEE/CVF WACV 2024** ([Paper](#), [GitHub](#))
6. **Laplacian-former: Overcoming the limitations of vision transformers in local texture detection**
Reza Azad, **Amirhossein Kazerooni**, Babak Azad, Ehsan Khodapanah Aghdam, Yury Velichko, Ulas Bagci, Dorit Merhof
Published in **MICCAI 2023 (top 14%)** ([Paper](#), [GitHub](#))
7. **Unlocking Fine-Grained Details with Wavelet-Based High-Frequency Enhancement in Transformers**
Reza Azad, **Amirhossein Kazerooni**, Alaa Sulaiman, Afshin Bozorgpour, Ehsan Khodapanah Aghdam, Abin Jose, Dorit Merhof
Published in **MLMI @ MICCAI 2023** ([Paper](#), [GitHub](#))
8. **DermoSegDiff: A Boundary-Aware Segmentation Diffusion Model for Skin Lesion Delineation**
Afshin Bozorgpour*, Yousef Sadegheih*, **Amirhossein Kazerooni***, Reza Azad, Dorit Merhof
Published in **PRIME @ MICCAI 2023** ([Paper](#), [GitHub](#))
9. **Self-supervised Semantic Segmentation: Consistency over Transformation**
Sanaz Karimijafarbigloo, Reza Azad, **Amirhossein Kazerooni**, Yury Velichko, Ulas Bagci, Dorit Merhof
Published in **CVAMD @ IEEE/CVF ICCV 2023** ([Paper](#), [GitHub](#))
10. **Implicit Neural Representation in Medical Imaging: A Comparative Survey**
Amirali Molaei, Amirhossein Aminimehr, Armin Tavakoli, **Amirhossein Kazerooni**, Bobby Azad, Reza Azad, Dorit Merhof
Published in **CVAMD @ IEEE/CVF ICCV 2023** ([Paper](#), [GitHub](#))
11. **HiFormer: Hierarchical Multi-scale Representations Using Transformers for Medical Image Segmentation**
Moein Heidari*, **Amirhossein Kazerooni***, Milad Soltany*, Reza Azad, Ehsan Khodapanah Aghdam, Julien Cohen-Adad, Dorit Merhof
Published in **IEEE/CVF WACV 2023** ([Paper](#), [GitHub](#))
12. **Reducing Uncertainty in 3D Medical Image Segmentation under Limited Annotations through Contrastive Learning**
Sanaz Karimijafarbigloo, Reza Azad, **Amirhossein Kazerooni**, Dorit Merhof
Published in **MIDL 2024** ([Paper](#), [GitHub](#))
13. **MMCFormer: Missing Modality Compensation Transformer for Brain Tumor Segmentation**
Sanaz Karimijafarbigloo, Reza Azad, **Amirhossein Kazerooni**, Dorit Merhof
Published in **MIDL 2023 (Oral presentation)** ([Paper](#), [GitHub](#))
14. **MS-Former: Multi-scale Self-guided Transformer for Medical Image Segmentation**
Sanaz Karimijafarbigloo, Reza Azad, **Amirhossein Kazerooni**, Dorit Merhof
Published in **MIDL 2023 (Oral presentation)** ([Paper](#), [GitHub](#))
15. **An Intelligent Modular Real-Time Vision-Based System for Environment Perception**
Amirhossein Kazerooni, Amirhossein Heydarian, Milad Soltany, Aida Mohammadshahi, Abbas Omid, Saeed Ebadollahi
Published in **ML4AD @ NeurIPS 2022** ([Paper](#), [GitHub](#), [Workshop Page](#))

Journal Papers

1. **MedScale-Former: Self-guided multiscale transformer for medical image segmentation**
Sanaz Karimijafarbigloo, Reza Azad, **Amirhossein Kazerooni**, Dorit Merhof
Published in **Medical Image Analysis Journal (2025)** ([Paper](#))
2. **Diffusion models in medical imaging: A comprehensive survey**
Amirhossein Kazerooni, Ehsan K. Aghdam, Moein Heidari, Reza Azad, Mohsen Fayyaz, Ilker Hacıhaliloglu, Dorit Merhof
Published in **Medical Image Analysis Journal (2023)** ([Paper](#), [GitHub](#))

3. **Advances in Medical Image Analysis with Vision Transformers: A Comprehensive Review**
Reza Azad, **Amirhossein Kazerooni**, Moein Heidari, Ehsan Khodapanah Aghdam, Amirali Molaei, Yiwei Jia, Abin Jose, Rijo Roy, Dorit Merhof
Published in **Medical Image Analysis Journal** (2023) ([Paper](#), [GitHub](#))

Pre-print Papers

1. **AVIS: Adaptive Test-Time Scaling for Vision-Language Models**
Ahmadreza Jeddi*, Minh Ngoc Le*, **Amirhossein Kazerooni***, Hakki Can Karaimer, Hue Nguyen, Iqbal Mohomed, Michael Brudno, Konstantinos G. Derpanis, Alex Levinshstein, Babak Taati, Radek Grzeszczuk
Published on arXiv preprint arXiv:2606.11576 (2026) ([Samsung Research Toronto AI Center Internship](#))

2. **BurstGP: Enhancing Raw Burst Image Super Resolution with Generative Priors**
Dong Huo, Tristan Aumentado-Armstrong, Samrudhdi B. Rangrej, Maitreya Suin, Angela Ning Ye, Zhiming Hu, Amanpreet Walia, **Amirhossein Kazerooni**, Konstantinos G. Derpanis, Iqbal Mohomed, Alex Levinshstein
Published on arXiv preprint arXiv:2604.23508 (2026) ([Paper](#)) ([Samsung Research Toronto AI Center Internship](#))

3. **Enhancing Efficiency in Vision Transformer Networks: Design Techniques and Insights**
Moein Heidari, Reza Azad, Sina Ghorbani Kolahi, René Arimond, Leon Niggemeier, Alaa Sulaiman, Afshin Bozorgpour, Ehsan Khodapanah Aghdam, **Amirhossein Kazerooni**, Ilker Hacihaliloglu, Dorit Merhof
Published on arXiv preprint arXiv:2403.19882 (2024) ([Paper](#))

4. **Ensemble Neural Representation Networks**
Milad Soltany Kadarvish*, Hesam Mojtahedi*, Hossein Entezari Zarch*, **Amirhossein Kazerooni***, Alireza Morsali, Azra Abtahi, Farokh Marvasti
Published on arXiv preprint arXiv:2110.04124 (2021) ([Paper](#), [GitHub](#))

Honors & Awards

2024-present	Vector Research Grant , \$6000 per year	Toronto, Canada
2022	Ranked 3rd among 41 students who chose Control as a subfield	IUST, Iran
2022	Ranked 4th among 127 Electrical Engineering students	IUST, Iran
2021	Ranked 1st team in the national Rahneshan competitions for autonomous vehicles	INEF, Iran
2021	Ranked 2nd team in FIRA World Cup Competitions in Autonomous Cars League	FIRA, Iran
2015	Ranked 1st team in the A-lympiad National Mathematical Competition	Iran
2016	Hold a diploma from the A-lympiad World Mathematical Competition	Utrecht University, Netherlands
2017	Ranked within the top 1% among approximately 148,000 participants in the National University Entrance Exam	Iran

Skills

Programming	Python (primary), C/C++, MATLAB
Frameworks	PyTorch, JAX, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib
Tools	Linux, Git

Teaching Experience

- CSC2503: Foundations of Computational Vision, University of Toronto, Canada
- CSC420: Introduction to Image Understanding, University of Toronto, Canada (3 times)
- CSC384: Introduction to Artificial Intelligence, University of Toronto, Canada
- CSC148: Introduction to Computer Science, University of Toronto, Canada

Working Experience

Co-founder	Tehran, Iran
AIR (Artificial Intelligence and Robotics) Center	2020 - 2022
• Taught and mentored students in deep learning, machine learning, and Python.	

Professional Services

CVPR	Conference Reviewer for CVPR 2026
NeurIPS	Conference Reviewer for NeurIPS 2024, 2025
MICCAI	Conference Reviewer for MICCAI 2024
Media	Journal Reviewer for Medical Image Analysis Journal
EAAI	Journal Reviewer for Engineering Applications of Artificial Intelligence